RTR-Exercise: WebGL-1

Pts	Task
[8]	Familiarize yourself with the `BufferGeometry` class. Then derive new classes for a triangle, a quad, a circle, and a star and call them `TriangleGeometry, QuadGeometry, CircleGeometry,` and `StarGeometry`. Use these classes at least once in your code. *Hint: we looked at something similar in IPR already!*
[4]	Cleanup the implementation such that mostly OOP concepts are used. Make sure to use the `WebGLProgram` class so that you can load and use shaders in your application. Replace all calls to `gl.createProgram`, `gl.useProgram`, and `gl.uniform*.` `WebGLProgram` provides a `constructor`, a `use`, and `setUniform` functions instead. Furthermore, make sure only to use the `BufferGeometry` class (and any subclass of it). Remove unnecessary variables and functions in the JavaScript code (e.g., `createBuffer, createIndex, vs., fs, …`).
[8]	Design and implement a set of sports balls (or equipment) on the WebGL canvas. One object should be composed of multiple 2D shapes (e.g., rectangles, triangles, circles, lines, …). Pick at least 3 objects of your choice or design one yourself. Note: simplify or leave out too complex objects (i.e. leave out text). Create a sketch showing placement and sizes before starting the implementation (Hint: you are in NDC space). The sketch must be submitted electronically (i.e. submitted on Github)! The following picture shows some inspirations for balls:  (image credit: https://thehungryjpeg.com/product/3871498-sport-balls-soccer-tennis-volleyball-baseball-and-football-equipme) You can use a fully object-oriented approach (i.e., derive further classes from `BufferGeometry`) or a more functional approach (using Buffers, Triangles, and Quads). You can use transformations (it will make things more reusable), but it is not a requirement. Index buffers are recommended but not a must.
[2]	Use at least four different colors.
[2]	One object should use a clearly visible color gradient (e.g., a transition from green on one side to red on the other side of a rectangle/triangle). *Hint: you must use color attributes (buffer arrays) for this.*

GitHub Classroom: https://classroom50.org/mtd-rtr/rtr3-ws2026/assignments/webgl-01/accept
Deadline: Monday, 16 Nov 2026, 23:59